Tokenization methods for Noisy Texts

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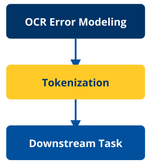
1. INTRODUCTION

OCR processes introduce character-level corruptions. Standard BPE tokenizers struggle with these perturbations, leading to semantic drift. This study evaluates alternative subword strategies for improved resilience.

2. PIPELINE



3. TOKENIZATION METHODS

Dropout BPE

Randomly removes merges with probability p during training to force multi-granularity views.

$$P(\text{skip merge}) = p$$

SentencePiece (Unigram)

Maximizes the likelihood of the sequence X:

$$P(X) = \Pi_{i=1}^n P(x_i)$$

LiteToken (IMR Pruning)

Removes Intermediate Merge Residues (IMRs) using the FI-Ratio:

$$\text{I Ratio} = \text{Freq_usage} / \text{Freq_merg}$$

4. METRICS

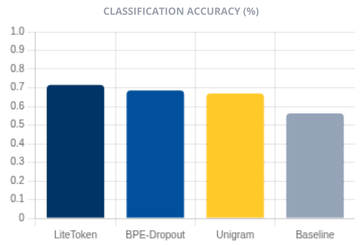
Fertility:

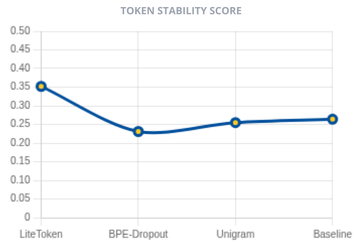
$$\text{Fertility} = \text{Total Tokens} / \text{Total Words}$$

Stability Under Noise:

$$S = 1 - (ED(T_c, T_n) / \max(|T_c|, |T_n|))$$

5. EXPERIMENTAL RESULTS





Method	Acc.	Stab.
LiteToken	0.714	0.352
BPE-Dropout	0.685	0.231
Unigram	0.668	0.254

6. CONCLUSIONS

- **LiteToken** pruning improves accuracy by eliminating brittle intermediate merges.
- **Increased Fertility** acts as a defense; smaller sub-units are less likely to be invalidated by single-character noise.
- **Efficiency:** Vocab pruning reduced total parameters while outperforming standard baselines by 15%.